

□ PRD — Food Delivery & Ordering Platform

1. Product Overview

Product Name

QuickBite

Vision

Build a scalable, multi-region food ordering and delivery platform connecting customers, restaurants, delivery agents, and admins, with strong consistency for money and orders, and clear operational ownership.

2. User Types & Roles

2.1 End User (Customer)

- Browse restaurants
 - View menus
 - Place orders
 - Choose payment method (Online / Cash on Delivery)
 - Track order & delivery status
 - View order history
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2.2 Restaurant (RBAC REQUIRED)

Roles inside a restaurant

1 ☐ Restaurant Owner

Permissions

- Manage restaurant profile
- Manage menu & pricing
- View all orders
- View financial balance
- View payout history
- Assign / revoke staff roles

2 ☐ Restaurant Manager

Permissions

- Manage menu
- Accept / reject orders
- Update order status
- View orders & basic analytics

3 ☐ Restaurant Staff

Permissions

- View incoming orders
- Update preparation status only

RBAC is enforced **per restaurant**, not globally.

2.3 Delivery Agent

- Accept or reject delivery tasks
 - Pick up orders from restaurants
 - Update delivery status:
 - Assigned
 - Picked up
 - Delivered
 - View delivery history
 - View earnings (read-only)
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2.4 Admin

- Manage restaurants
 - Manage delivery agents
 - View all orders
 - Monitor payments
 - Record restaurant payouts
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3. Order Lifecycle (End-to-End)

1. Customer places order
2. Payment decision:
 - **Online payment** → payment authorization required
 - **Cash on Delivery (COD)** → no pre-payment

3. Order sent to restaurant
 4. Restaurant accepts or rejects
 5. Order prepared
 6. Delivery agent assigned
 7. Order picked up
 8. Order delivered
 9. Financial settlement recorded
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4. Delivery Assignment

Assignment Model

- **Automatic assignment** (default)
 - Based on proximity
 - Agent availability
- **Manual override** (Restaurant)
 - Reassign agent
 - Handle edge cases

Delivery agents can:

- Accept
 - Reject (limited retries)
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5. Payments & Money Flow

Payment Methods

- Online payments
 - Cash on Delivery (COD)
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Online Payment Flow

- Payment authorized before order confirmation
 - Idempotent payment handling
 - Order created only after confirmed payment
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COD Flow

- Order created without payment
 - Delivery agent collects cash
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Restaurant Balance Model

- Each restaurant has a **running balance**
 - Balance increases when:
 - Order is delivered successfully
 - Platform commission deducted automatically
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Payouts

- Restaurants are paid externally (bank transfer)
 - **Admin records payout events in the system**
 - Balance reduced after payout is recorded
 - Full payout history retained
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6. Functional Requirements

Customer

- Account management
 - Restaurant discovery
 - Menu browsing
 - Cart management
 - Order placement
 - Payment selection
 - Order tracking
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Restaurant

- Menu CRUD
- Order management
- Status updates
- Financial balance view

- Role management (Owner only)
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Delivery Agent

- Task assignment
 - Status updates
 - Delivery history
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Admin

- Full visibility across system
 - Manual overrides
 - Financial reconciliation
 - Payout recording
 - Manage restaurants
 - Reports
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7. Non-Functional Requirements

- **Availability:** High (order placement & payments are critical)
- **Latency:**
 - Browsing <1s
 - Checkout < 1-3s
- **Consistency:**

- Strong consistency for orders & payments
 - Eventual consistency acceptable for analytics
 - **Scalability:** Must handle peak traffic
 - **Security:**
 - Strong authorization boundaries
 - Secure payment handling
 - **Multi-Region:**
 - Active traffic in multiple regions
 - Regional reads
 - Strong consistency where required
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8. Capacity & Scale (Back-of-the-Envelope)

Traffic

- **MAU:** 1,000,000
- **DAU:** 200,000
- **Actions / user / day:** 27
- **Requests / day:** ~5.4M
- **Average RPS:** ~62
- **Peak RPS:** ~620

Read / Write Ratio

- **Reads:** 85%

- **Writes:** 15%
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Storage

- **Orders growth:** ~146 GB / year

Bandwidth

- **~270 GB / day**
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9. Data Retention

- Orders: retained long-term, we show in the customer app and restaurant only current year orders. Others are stored in another database storage
 - Payments, same as orders
 - Logs: time-limited (30 days)
 - Analytics data: aggregated & archived
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10. Multi-Region Assumptions

- Users served from nearest region
 - Restaurants tied to a primary region
 - Orders processed in restaurant's region
 - Financial data requires strong consistency
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11. Constraints & Risks

- Meal-time traffic spikes
 - Payment provider failures
 - Delivery agent shortages
 - Cross-region consistency complexity
 - Settlement reconciliation errors
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12. Success Criteria

- Orders complete end-to-end without loss
 - Payments are correct and auditable
 - Restaurant balances are accurate
 - System remains stable under peak load
 - Clear ownership and authorization boundaries
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13. Out of scope

- Recommendation systems
- Dynamic pricing
- Loyalty programs
- AI-based delivery optimization
- Reviews